Homework #6. Due on Thursday, Oct 15th, 11:59pm on Canvas

Reading for this homework: Sections 3.1 and parts of 3.4.

Plan for next week (Oct 12, 14): 3.2 (the rank of a matrix), 4.2 (determinants of order n), possibly start 4.3 (properties of determinants). Note: my plan is to define the determinants by the Leibniz formula `https://en.wikipedia.org/wiki/Leibniz_formula_for_determinants` instead of the cofactor expansion used in the book.

Problems:

For the first 2 problems do not assume anything about the rank except the following definitions:

(1) Let $T : V \to W$ be a linear map where V and W are both finite-dimensional. The rank of T, denoted by $\mathrm{rk}(T)$, is defined by

$$\mathrm{rk}(T) = \dim(\mathrm{Im}(T)) = \dim(T(V)).$$

(2) If F is a field and $A \in \mathrm{Mat}_{m\times n}(F)$, we define its rank $\mathrm{rk}(A)$ by $\mathrm{rk}(A) = \mathrm{rk}(L_A)$ where $L_A : F^n \to F^m$ is the map of left multiplication by A, that is, $L_A(v) = Av$ for all $v \in F^n$.

Problem 1: Let $T : V \to W$ and $S : V \to W$ be linear maps.

(a) Prove that $\mathrm{Im}(T+S) \subseteq \mathrm{Im}(T)+\mathrm{Im}(S)$ and give an example showing that the equality may not hold.

(b) Now assume that V and W are finite-dimensional. Prove that

$$\mathrm{rk}(T + S) \leq \mathrm{rk}(T) + \mathrm{rk}(S).$$

Problem 2: Let $T : V \to W$ be a linear map and Z be a subspace of W.

(a) Prove that $T^{-1}(Z)$ is a subspace of V. Here $T^{-1}(Z)$ is the preimage of Z, that is, $T^{-1}(Z) = \{v \in V : T(v) \in Z\}$.

(b) Assume that V and W are finite-dimensional. Prove that

$$\dim(T^{-1}(Z)) \leq \dim(Z) + \dim(\mathrm{Ker}(T))$$

Hint: Apply the rank-nullity theorem to a suitable map obtained from T by changing the domain.

Problem 3: In parts (a) and (b) let U, V, W be finite-dimensional vector spaces and $T : U \to V$ and $S : V \to W$ linear maps.

(a) Prove that

$$\mathrm{null}(ST) \leq \mathrm{null}(S) + \mathrm{null}(T)$$

where by definition $\text{null}(R) = \dim(\text{Ker}(R))$ for a linear map R. **Hint:** You can express $\text{Ker}(ST)$ in a suitable way using preimages, so that the result would follows immediately from Problem 2.

(b) Prove that $\text{rk}(ST) \geq \text{rk}(S) + \text{rk}(T) - \dim(V)$.

(c) State and prove the analogue of (b) dealing with ranks of matrices.

Problem 4: Let F be a field and $m, n \in \mathbb{N}$. Recall from class that

(i) $\mathcal{E}_{i,j}(\lambda)$ is the elementary row operation on the set $\text{Mat}_{m\times n}(F)$ which adds the jth row multiplied by λ to the ith row and does not change any other row (symbolically $\text{row}_i \mapsto \text{row}_i + \lambda\,\text{row}_j$).

(ii) $\mathcal{E}'_{i,j}(\lambda)$ is the elementary column operation which adds the ith column multiplied by λ to the jth column and does not change any other column ($\text{col}_j \mapsto \text{col}_j + \lambda\,\text{col}_i$).

(iii) $E_{ij}(\lambda)$ is the elementary (square) matrix (of suitable size) which has 1's on the diagonal, λ in the (i, j)-position and 0 everywhere else. Thus, $E_{ij}(\lambda) = I_k + \lambda e_{i,j}$ where k is the size of the matrix and I_k is the identity $k \times k$ matrix.

Prove (part of) Lemma 11.3 from class: for any $A \in \text{Mat}_{m\times n}(F)$ we have

(a) $(\mathcal{E}_{i,j}(\lambda))(A) = E_{i,j}(\lambda)A$ and

(b) $(\mathcal{E}'_{i,j}(\lambda))(A) = AE_{i,j}(\lambda)$,

that is, applying $E_{i,j}(\lambda)$ (resp. $\mathcal{E}'_{i,j}(\lambda)$) is the same as multiplying by $E_{i,j}(\lambda)$ on the left (resp. on the right).

Note: The notation $E_{i,j}(\lambda)$ is slightly ambiguous as it does not specify the size of the matrix. In particular, if $m \neq n$, then $E_{i,j}(\lambda)$ denotes matrices of different sizes in (a) and (b) above.

Problem 5: Recall two other types of elementary transformations introduced in class: $\mathcal{F}_{i,j}$ (swapping the ith and j^{th} rows) and $\mathcal{D}_i(\lambda)$ (multiplying the ith row by some nonzero $\lambda \in F$). Prove that $\mathcal{F}_{i,j}$ can expressed as a composition of 3 operations of the form $\mathcal{E}_{k,l}(\lambda)$ and 1 operation of the form $\mathcal{D}_k(\lambda)$ (in a suitable order and for suitable k, l and λ).

Problem 6: In Lecture 11 we explained why every matrix A can be reduced to a matrix R in RREF (reduced row echelon form) using elementary row operations. The ultimate goal of this problem is to prove that such a matrix R is unique. Given $A \in \text{Mat}_{m\times n}(F)$, we define its rowspace $\text{Rowspace}(A)$ to be the subspace of F^n spanned by the rows of A.

(1) Suppose that $A, B \in \mathrm{Mat}_{m\times n}(F)$ are row equivalent (that is, can be obtained from each other by a sequence of elementary row operations). Prove that $\mathrm{Rowspace}(A) = \mathrm{Rowspace}(B)$. **Note:** The converse of (1) is also true, but requires more work to prove.

Now let $R \in \mathrm{Mat}_{m\times n}(F)$ be a matrix in RREF. Our goal is to show that R can be recovered from its rowspace $\mathrm{Rowspace}(R)$. Together with (1) this implies that any A is row equivalent to a UNIQUE matrix in RREF.

(2) Let $e_1, \dots e_n$ be the standard basis of F^n (which is viewed here as the space which naturally contains $\mathrm{Rowspace}(R)$). For each $1 \le k \le n$ let $V_k = \mathrm{Span}(e_k, \dots, e_n)$ be the span of all elements of the standard basis starting with e_k. Also set $V_{n+1} = \{0\}$. Thus $V = V_1 \supset V_2 \supset \cdots \supset V_n \supset V_{n+1} = \{0\}$. Let $W_k = V_k \cap \mathrm{Rowspace}(R)$ for $1 \le k \le n+1$.

Now take $1 \le k \le n$. Prove that $W_k \ne W_{k+1} \iff R$ has a pivot in the kth column. Thus, the set of columns of R which contain pivots is determined by $\mathrm{Rowspace}(R)$.

(3) Let $i_1 < i_2 < \ldots < i_p$ be the indices of columns of R which contain pivots. Note that since R is in RREF, this means that the first p rows of R are nonzero, and the remaining rows are zero.

Let $T_0 : F^n \to F^p$ be the map which simply removes the jth component for each $j \notin \{i_1, \dots, i_p\}$ (and preserves the order of the remaining components). For example, if $n = 5$, $p = 2$, $i_1 = 2$ and $i_2 = 4$, we have $T_0((a_1, a_2, a_3, a_4, a_5)) = (a_2, a_4)$.

Let $T : \mathrm{Rowspace}(R) \to F^p$ be the restriction of T_0 to $\mathrm{Rowspace}(R)$. Prove that T is an isomorphism and for all $1 \le i \le p$ we have $T(\mathrm{row}_i(R)) = \varepsilon_i$, where $\{\varepsilon_1, \dots, \varepsilon_p\}$ is the standard basis of F^p. Thus, $\mathrm{row}_i(R)$ is the unique element of $\mathrm{Rowspace}(R)$ which is mapped to ε_i under T. Since the indices $i_1, \dots, i_p$ (and hence also the number p) are determined by $\mathrm{Rowspace}(R)$ by (2) and all rows of R below the pth row are zero, it follows that each row of R can be recovered from $\mathrm{Rowspace}(R)$, as desired.

Problem 7: Let $A = \begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix}$. Reduce A to the identity matrix only using operations of the form $\mathcal{E}_{12}(\lambda)$ and $\mathcal{E}_{21}(\lambda)$ for some integer λ. Then use Problem 4 to explicitly write A as a product of matrices of the form $E_{12}(\lambda)$ and $E_{21}(\lambda)$.

Problem 8: Let $n \in \mathbb{N}$ and F a field.

(a) Let $A \in \mathrm{Mat}_n(F)$ be an invertible matrix. Prove that the transposed matrix A^t is also invertible and $(A^t)^{-1} = (A^{-1})^t$.

(b) Let $A \in \mathrm{Mat}_n(F)$ be such that $A^k = 0$ for some $k \in \mathbb{N}$ (such matrices are called *nilpotent*). Prove that the matrix $B = I_n + A$ is invertible (as before, I_n is the identity matrix) and find an explicit formula for the inverse of A.

Do not use any facts about inverses except the definition. In (b) you may want to start with $k = 2$ (possibly followed by $k = 3$) before figuring out the general formula.